

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:20

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I Arunprasath R(192524077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Arunprasath R

Annexure A

CO1 Assessment: Constraint-Based Problem Solving

9. Missionaries and Cannibals (State-Space Search under Safety Constraints)

Three missionaries and three cannibals are on the left bank of a river with a boat that carries at most two people. The boat cannot cross empty. On either bank, the missionaries must never be outnumbered by cannibals (if any missionaries are present).

- i. Formulate the problem: state representation, initial state, actions, transition model, goal test and path cost.
- ii. Identify which states violate the safety constraint and must be pruned.
- iii. Draw the state space and solve the problem using Breadth-First Search.
- iv. Give the minimum number of crossings and justify why BFS guarantees this minimum.

ANS:

i. Problem Formulation

- **State Representation:** A state is represented as a triple (M_left, C_left, Boat_side) where:
 - M_left = number of missionaries on the left bank
 - C_left = number of cannibals on the left bank
 - Boat_side = L (boat on left) or R (boat on right) Since there are 3 missionaries and 3 cannibals total, the right bank values can be inferred: $M_{right} = 3 - M_{left}$, $C_{right} = 3 - C_{left}$.
- **Initial State:** (3,3,L) → all on left bank, boat on left.
- **Actions:** Possible moves are:
 - One missionary crosses
 - Two missionaries cross
 - One cannibal crosses
 - Two cannibals cross
 - One missionary + one cannibal cross The boat must carry at least one person and at most two.

- **Transition Model:** If boat is on left, subtract from left and add to right. If boat is on right, subtract from right and add to left. Example: $(3,3,L)$ with action “two cannibals cross” $\rightarrow (3,1,R)$.
- **Goal Test:** $(0,0,R) \rightarrow$ all missionaries and cannibals safely on right bank.
- **Path Cost:** Each boat crossing = cost 1. The objective is to minimize crossings.

ii. Safety Constraint

- Rule: On either bank, if missionaries > 0 , then missionaries $\geq$ cannibals.
- Unsafe states must be pruned. Examples:
 - $(2,3,L) \rightarrow$ left bank has 2 missionaries, 3 cannibals (unsafe).
 - $(1,2,R) \rightarrow$ right bank has 1 missionary, 2 cannibals (unsafe).
- These states are excluded from the search tree.

iii. State Space and BFS Solution

Breadth-First Search explores states level by level, ensuring shortest solution.

Solution Path (one optimal sequence):

1. $(3,3,L) \rightarrow (3,1,R)$ [two cannibals cross]
2. $(3,2,L)$ [one cannibal returns]
3. $(3,0,R)$ [two cannibals cross]
4. $(3,1,L)$ [one cannibal returns]
5. $(1,1,R)$ [two missionaries cross]
6. $(2,2,L)$ [one missionary + one cannibal return]
7. $(0,2,R)$ [two missionaries cross]
8. $(0,3,L)$ [two cannibals return]
9. $(0,1,R)$ [two cannibals cross]
10. $(0,2,L)$ [one cannibal returns]
11. $(0,0,R)$ [final two cannibals cross]

iv. Minimum Crossings

- BFS guarantees optimality because it explores all states at depth d before $d+1$.
- Minimum crossings = **11**.
- Any shorter path would have been discovered earlier in BFS, so this is provably minimal.

10. Sensorless Corridor Robot (Search with Partial Information)

A cleaning robot is placed in a corridor of four cells numbered 1 to 4 from left to right. It has no sensors, so it does not know which cell it starts in. The action Right moves it one cell to the right (it stays in cell 4 if already there), and Left moves it one cell to the left (it stays in cell 1 if already there). The robot must be certain that it is in cell 4.

- i. Define the initial belief state and the size of the belief-state space.
- ii. Show the belief state after each action in the sequence [Right, Right, Right].
- iii. Prove that no shorter action sequence guarantees reaching cell 4.
- iv. Suppose the robot gains a bump sensor that reports when it tries to move into a wall. Write a contingency plan that reaches cell 4 and compare it with the sensorless plan.

ANS:

i. Initial Belief State

- Corridor has 4 cells: {1,2,3,4}.
- Robot has no sensors, so initial belief state = {1,2,3,4} (could be anywhere).
- **Belief-state space size** = $2^4 - 1 = 15$ (all non-empty subsets of {1,2,3,4}).

ii. Belief State After [Right, Right, Right]

- Start: {1,2,3,4}
- After 1st Right: {2,3,4} (if at 1 → now at 2; if at 2 → now at 3; if at 3 → now at 4; if at 4 → stays at 4).
- After 2nd Right: {3,4}
- After 3rd Right: {4} → robot is certain it is in cell 4.

iii. Proof of Minimality

- After 2 moves: belief = {3,4} → ambiguity remains.
- Only after 3 moves does belief collapse to {4}.
- Therefore, **3 actions is the minimum** sequence to guarantee certainty.
- Any shorter sequence cannot eliminate uncertainty.

iv. With Bump Sensor

- Robot gains a bump sensor that detects when it tries to move into a wall.
- **Contingency Plan:**
 - Move Right until bump detected.
 - If bump occurs immediately → robot was already at cell 4.
 - If bump occurs later → robot knows it has reached cell 4.
- **Comparison:**
 - Sensorless plan = fixed sequence [Right, Right, Right] (always 3 moves).
 - With bump sensor = adaptive plan, potentially fewer moves:
 - If starting at 4 → only 1 move needed.
 - If starting at 3 → 1 move needed.
 - If starting at 2 → 2 moves needed.
 - If starting at 1 → 3 moves needed.

- Thus, bump sensor improves efficiency by reducing unnecessary moves and provides certainty faster in some cases.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided